The person can travel from A to B in 2 ways and the person can travel from B to C in 3 ways.

By the Fundamental Principle of multiplication, the person can travel from A to C simultaneously in $2 \times 3 = 6$ ways.

[Note: Observe the answer part and the diagram carefully.]

Example 7.3

A company allots a code on each different product they sell. The code is made up of one English letter and two digit numbers. How many different codes are possible?

Solution:

There are 26 English Letters (A to Z) and other two digit numbers (0 to 9) are given.

Letter	Number	Number
26 ways	10 ways	10 ways

The letter place can be filled in 26 ways with the 26 alphabets *A* to Z.

The ten's place can be filled in 10 ways with the digits 0 to 9.

The unit's place also can be filled in 10 ways with the digits 0 to 9.

So the number of product codes can be formed in $26 \times 10 \times 10$ ways $= 2600$ ways.

Example 7.4

How many four digit numbers can be formed by using the digits 2, 5, 7, 8, 9, if the repetition of the digits is not allowed?.

Solution:

Thousand	Hundred	Ten	One
5 ways	4 ways	3 ways	2 ways

The thousand's place can be filled with the 5 digits in 5 ways.

Since the repetition is not allowed, the hundred's place can be filled with the remaining 4 ways.

Similarly, for the ten's place can be filled with the remaining 3 digits in 3 ways and the unit's place can be filled with the remaining 2 digits in 2 ways.